

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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ABSTRACT

Human evaluation remains essential for ranking generative models, but it is expensive, noisy, and sensitive to differences in rater expertise, attention, and diligence. Pairwise comparisons make the evaluation task easier for participants, yet they still require an aggregation method that can separate item quality from rater reliability. Existing Bradley-Terry-based approaches either treat all raters as equally reliable or model rater variability with optimization procedures that lack the convergence guarantees needed for stable, reviewer-facing leaderboards. We contribute BBQ, a Bayesian Bradley-Terry model that makes rater quality an explicit part of the aggregation problem. The method downweights unreliable judgments through interpretable rater-quality parameters and derives closed-form Expectation-Maximization updates with guaranteed monotonic likelihood improvement. Across language and image-compression preference datasets, including a newly collected inhomogeneous rater quality dataset, BBQ produces robust rankings, faster convergence, and well-calibrated uncertainty estimates relative to Bradley-Terry baselines. Together, these contributions provide a practical framework for more reliable and cost-effective human evaluation of generative models.

1 INTRODUCTION

The central evaluation problem for generative models is that automatic metrics only partially capture what people perceive as high quality. This gap is especially visible for large language models (LLMs) and image generators: BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) provide limited evidence for LLM quality, while PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) only partly reflect perceived image quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). Human preference data therefore remains a core ingredient for meaningful model rankings. This is why leaderboards and comparative studies for both large language models and Learned Image Compression (LIC) methods rely heavily on human judgments.

The practical obstacle is that human evaluation is costly, time-consuming, and noisy. Pairwise comparison protocols reduce this burden by asking participants to choose between two items instead of assigning absolute scores (Zerman et al., 2018; Wang et al., 2023). These choices are easier for raters and often more consistent, but they do not by themselves solve the aggregation problem. Exhaustively collecting all pairs is infeasible because the number of pairs grows quadratically with the number of items. When only a sparse subset of comparisons is available, win rates are not reliable global quality estimates because each win rate depends on the strength of the opponents an item happened to face.

Thus, the first contribution required by a useful human-evaluation pipeline is robust aggregation: sparse pairwise choices must be converted into an overall ranking that can support leaderboard decisions. The Bradley-Terry (BT) model (Bradley & Terry, 1952) is a standard solution because it updates item scores from comparison outcomes in an interpretable probabilistic framework. Its broad use across domains makes it a natural foundation for preference aggregation, but also makes its limitations consequential for modern model evaluation.

Standard BT aggregation leaves two reviewer-relevant gaps. First, estimation quality depends on the chosen optimization procedure: iterative scaling (Ford Jr, 1957), gradient descent, and minorization maximization (MM) (Hunter, 2004) have different stability properties. MM guarantees convergence

to the global maximum for the basic BT model, while widely used gradient-based procedures related to Elo ratings do not provide the same guarantee (Hunter, 2004). For extended BT models with home-field effects (Agresti, 2010), multiple-comparison structure (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), even MM algorithms may reach only local optima (Hunter, 2004). Second, the standard model treats human judgments as equally reliable, although participants may rush, guess, lose focus, or differ in expertise. Ignoring this rater heterogeneity can let unreliable judgments distort item scores and reduce the stability of the final ranking.

We address these gaps with BBQ, a Bayesian Bradley–Terry variant that jointly estimates item quality and rater reliability. The methodological contribution is twofold: BBQ introduces rater-specific quality parameters that control how much each comparison influences the ranking, and it derives a latent-variable EM algorithm with closed-form updates and monotonic likelihood improvement. To the best of our knowledge, this combination of Bayesian BT modeling, explicit rater quality, and closed-form EM updates has not appeared in prior work. Compared with noisy-rater approaches based on gradient optimization, BBQ is designed to make convergence behavior, rater influence, and item uncertainty visible to the evaluator. The Bayesian formulation also regularizes item scores and provides uncertainty estimates, enabling rankings that are easier to interpret and compare across studies. As a result, partially reliable raters can still contribute useful signal, while inattentive or inconsistent participants receive reduced influence.

Empirically, we evaluate BBQ across multiple human preference datasets and introduce a newly collected *inhomogeneous rater quality (IHQ)* dataset¹. These experiments support three front-loaded claims: BBQ is robust to noisy raters, converges efficiently, and produces uncertainty estimates suitable for interpreting close ranking differences. The contribution is not limited to one modality; the same aggregation framework applies whenever noisy pairwise comparisons must be transformed into reliable global rankings. Overall, the paper contributes a practical path toward more cost-efficient, interpretable, and reproducible human studies for evaluating AI-generated content.

2 RELATED WORK

Prior work establishes pairwise comparison as a strong interface for human ranking studies. Asking participants to choose between two items, rather than score one item in isolation, improves sensitivity to subtle differences and reduces response variability (Zerman et al., 2018; Wang et al., 2023). This design choice underlies many machine-learning leaderboards: participants express preferences between alternatives, and an aggregation model converts those choices into a global ranking. Chatbot Arena (Chiang et al., 2024) and the CLIC image compression challenge (CLIC, 2025) are representative examples that use pairwise comparisons and Bradley-Terry (BT)-style aggregation.

The BT model (Bradley & Terry, 1952), originally introduced for sports rankings, contributes a probabilistic mapping from pairwise outcomes to item strengths. Because it estimates the probability that one item is preferred to another, it has become a basic tool for studies in games, consumer preferences, and other ranking applications. The literature has extended this foundation in several directions. The Plackett-Luce model expands BT from pairwise outcomes to rankings over multiple items (Plackett, 1975; Luce et al., 1959) by defining permutation probabilities through successive BT terms (Luce et al., 1959). Other extensions target domain-specific structure, including home-field advantages (Agresti, 2010), ties (Rao & Kupper, 1967), and group-level comparisons (Huang et al., 2006).

Optimization for BT models is another axis along which our contribution is positioned. Maximum likelihood estimation is the classical route for the basic BT model, and Zermelo (1929) introduced an iterative estimator that remains influential. Lange et al. (2000) later showed that this procedure is a minorization-maximization (MM) algorithm, which optimizes surrogate functions to reach a likelihood maximum. Hunter (2004) extended MM methods to generalized BT models and identified conditions under which they converge to the MLE. For the classical BT model the objective is convex and the global optimum is recovered (Hunter, 2004), but extensions involving home-field advantage, multiple comparisons, or ties may only guarantee convergence to a local maximum (Hunter, 2004).

Bayesian BT formulations contribute regularization and uncertainty quantification, both of which are important when comparisons are sparse or noisy (Adams, 2005; Guiver & Snelson, 2009; Caron

¹The IHQ dataset will be released upon acceptance.

& Doucet, 2012). Caron & Doucet (2012) showed that MM algorithms can be interpreted as Expectation-Maximization (EM) procedures, enabling Bayesian inference through Gibbs sampling. That construction yields tractable complete-data likelihoods and convergence properties for the resulting Markov Chain Monte Carlo methods. For our purposes, the key lesson is that Bayesian structure can smooth item-skill estimates and expose uncertainty, but existing formulations do not by themselves resolve heterogeneous rater quality.

Annotator-quality models address the specific assumption that limits standard BT aggregation: not every comparison is equally reliable. This issue is acute in crowdsourced evaluations, where participant quality can vary substantially (Chen et al., 2013). Chen et al. (2013) proposed a Bayesian model that jointly estimates item quality and rater reliability using annotator-specific parameters. Their method uses Gaussian priors for item and rater parameters and likelihood scaling factors to modulate each annotator’s influence. However, posterior inference is performed with gradient-based optimization, which can be difficult in high-dimensional spaces and does not provide convergence guarantees. Our contribution retains the rater-quality objective while replacing this optimization path with an EM procedure that guarantees monotonic likelihood improvement. The Elo-based rater model developed by Google Research implements a simplified version of this approach and is widely used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and related research (Mentzer et al., 2020; Ballé et al., 2025).

The gap left by this literature is therefore specific and actionable. To the best of our knowledge, no prior work combines a Bayesian Bradley–Terry formulation, explicit rater-quality parameters, and closed-form EM updates. Relative to Chen et al. (2013), our method replaces Gaussian-prior gradient optimization with conjugate priors and EM updates that ensure monotonic likelihood improvement. Relative to the Elo-style rater model used in CLIC (CLIC, 2025), BBQ provides a principled Bayesian treatment with uncertainty estimates and interpretable rater-quality parameters.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

Our model contribution starts from the standard problem of converting noisy pairwise comparisons into a reliable ranking. Let K items be compared in pairs by R raters, and let D denote the observed comparison outcomes. For a pair of items i and j , the Bradley-Terry model of Bradley & Terry (1952) assigns the win probability

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ is the skill rating of item $k \in \{1, 2, \dots, K\}$. This formulation is interpretable because item scores are inferred directly from observed wins and losses. Its limitation, and the limitation our method targets, is that every comparison is treated as equally reliable. Human evaluation often violates this assumption: raters differ in expertise, attention, and diligence, so their judgments can be unevenly informative.

BBQ makes this reliability dimension explicit by introducing a rater-specific quality parameter $q_r \in [0, 1]$. With probability q_r , rater r provides an informed judgment governed by the Bradley-Terry probability; with probability $1 - q_r$, the rater behaves as a random guesser who selects between the two items with equal probability. This yields the following mixture model for the probability that rater r ranks item i above item j :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The resulting observed-data log-likelihood is

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ is the number of comparisons between items i and j made by rater r , and $w_{r,ij}$ is the number of times that rater ranks item i above item j . This likelihood couples item skills and rater qualities, so a closed-form maximum-likelihood estimator for λ and q is not available. Gradient descent is possible but introduces learning-rate choices and lacks a convergence guarantee. Our second methodological contribution is therefore to estimate the model with an Expectation–Maximization (EM) algorithm, which avoids learning-rate tuning and guarantees monotonic improvement of the observed-data likelihood until convergence to a stationary point (Dempster et al., 1977).

3.2 THURSTONIAN INTERPRETATION

The EM derivation is made tractable by using the Thurstonian interpretation of Bradley-Terry from Caron & Doucet (2012) and Diaconis (1988). In this view, a comparison between items i and j is a race in which the items have random arrival times Y_i and Y_j . We assume these arrival times follow exponential distributions:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

The item with the smaller arrival time wins the comparison. This construction recovers the standard Bradley-Terry probability:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

To connect this interpretation to rater quality, BBQ introduces latent variables that expose whether each observed comparison is informative or random. The first latent variable is the indicator

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which records whether the c -th comparison of items i and j by rater r follows the Bradley-Terry component. Using these indicators, we define $Z_{r,ij}$ as the sum of the minimal arrival times across the $n_{r,ij}$ comparisons by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, this latent total follows a Gamma distribution,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape parameter α and inverse scale β . This latent-variable construction is the mechanism that keeps the rater-quality extension tractable: it preserves the BT race interpretation while enabling closed-form EM updates.

3.3 EXPECTATION-MAXIMIZATION UPDATES

The EM updates are the main algorithmic contribution of BBQ. They compute maximum a posteriori (MAP) estimates of the item skills λ and rater qualities q by treating the informative-comparison indicators and Thurstonian arrival-time quantities as latent variables. This formulation gives a practical ranking algorithm whose posterior objective improves monotonically and converges to a stationary point.

We place conjugate priors on both sets of parameters. Each item skill has a Gamma prior, $\lambda_k \sim \Gamma(a, b)$, which is conjugate to the exponential race model, and each rater-quality parameter has a Beta prior, $q_r \sim B(\alpha, \beta)$.

The algorithm alternates between two transparent steps: compute the expected complete-data log-posterior under the current estimates (λ^*, q^*) , and then maximize that expectation in closed form.

E-step: Expectation The E-step assigns responsibility to the informative-comparison component. Formally, it computes the expected complete-data log-posterior Q , the expectation of the log-posterior over observed and latent variables given the data and the previous parameter estimates:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} \left[\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q) \right]. \quad (9)$$

The complete-data log-likelihood ℓ_c decomposes into three terms, which separate the latent race times, the rater-quality indicators, and the observed comparisons:

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization The M-step converts those responsibilities into closed-form parameter updates. The key shared quantity is the posterior probability that a comparison from rater r came from the Bradley-Terry component rather than the random-guessing component. We denote this quantity by γ , and it functions as a data-dependent weight on each judgment: comparisons that agree with the current model receive more influence, while likely random comparisons receive less. It is given by:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the current Bradley-Terry probability that item i beats item j .

Thus, γ is the algorithmic link between the rater-quality model and the ranking: higher values mark comparisons that are more likely to be meaningful under the current parameters.

The rater-quality update is a regularized weighted average of meaningful judgments. Its numerator counts the effective number of non-random comparisons assigned to rater r , weighted by γ and regularized by the Beta prior; its denominator normalizes by the rater’s comparison count and prior mass. Consequently, raters whose decisions repeatedly align with the inferred item strengths receive higher quality estimates:

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons made by rater r .

The item-skill update uses the same responsibilities to aggregate evidence for each item. The numerator collects the effective wins for item i across raters, regularized by the Gamma prior; the denominator accounts for the comparison exposure of item i under the current skill estimates. This update is the closed-form mechanism by which BBQ turns weighted pairwise evidence into item scores:

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

Appendix B provides the full derivation of these EM updates.

4 EXPERIMENTAL RESULTS

The experiments are organized around the paper’s main claims: BBQ should produce stable rankings, remain robust when rater quality is heterogeneous, converge efficiently, and provide useful uncertainty estimates. We compare our **Bayesian Bradley-Terry** model with rater **Quality (BBQ)** against Bayesian Bradley-Terry (Bayes-BT) (Caron & Doucet, 2012) and a gradient-descent BT model with rater quality (Crowd-BT) (Chen et al., 2013). For Crowd-BT, we use the Google Research implementation (Google, 2025) used by CLIC (2025) and Ballé et al. (2025); the hyperparameters for all experiments are reported in Section C.

We test these claims on language and image preference datasets: HUMAINE (Team, 2025) and MT-Bench (Zheng et al., 2023) for large language models; WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), and ConHa (Aczel & Wattenhofer, 2024) for image compression; and our newly collected inhomogeneous rater quality (IHQ) dataset built from the CLIC2024 image test set (CLIC, 2025). The IHQ dataset is central to the rater-quality contribution because it is partitioned into screened and unscreened subsets, allowing us to compare high-quality, low-quality, and mixed rater pools. Details of IHQ collection appear in Section E, and dataset sizes, rater counts, item counts, and comparison counts are summarized in Section D and Table 2.

Because real datasets do not provide ground-truth rankings, our evaluation focuses on reproducibility of rankings under resampling. We first verified on simulated data that all methods recover the same ordering as the sample size grows. For crowdsourced datasets, we approximate the reference ranking by the ranking obtained from the full dataset and check whether each method recovers the same top item. All three models agree on the top-ranked item for every dataset except IHQ-unscreened. On IHQ-unscreened, Crowd-BT and Bayes-BT fail to identify the reference image as the highest-quality item, while BBQ succeeds, directly illustrating the value of modeling rater quality under noisy annotation.

A reviewer-facing aggregation method should also be stable: repeating the study should produce a similar ordering. We estimate this stability through bootstrapping, which measures ranking variability under resampled rater pools; implementation details are given in Section F. The WD and HiFiC results require caution because those studies used active pair selection rather than uniformly sampled comparisons. HiFiC used binary search, and WD selected pairs by maximum information gain, so their bootstrap variability is not directly comparable to uniformly sampled studies.

We report two complementary metrics. Top-1 agreement measures the fraction of bootstrap samples that select the same best item as the full dataset, which matters when the main decision is choosing a winner. Kendall’s Tau (τ) measures ordinal correlation between rankings (Kendall, 1938), capturing stability across the full ordered list.

4.1 PERFORMANCE ACROSS DATASETS

Table 1 summarizes the first empirical contribution: BBQ improves ranking accuracy and stability across heterogeneous evaluation settings. Datasets with more comparisons per model, such as MT-Bench, WD, and HiFiC, are generally easier for all methods. HUMAINE is an important exception, showing that comparison count alone does not determine stability; rater consistency and diversity also shape aggregation performance.

Across the full benchmark suite, BBQ is the most consistently stable method. It identifies the top-performing item most often, tying for the best Top-1 agreement on three datasets. It also achieves the best Kendall’s Tau on five of eight datasets and ranks second on the remaining three, supporting the claim that the rater-quality extension improves robustness without sacrificing performance in cleaner settings.

The cases where BBQ ranks second in Kendall’s Tau are also informative about the method’s scope. They correspond to high-quality, homogeneous rater sets: all methods recover the top item on MT-Bench; WD, collected by Ballé et al. (2025), contains only five raters who are likely the paper’s authors; and IHQ-screened includes only raters who passed quality filters. When raters are already consistently reliable, explicitly modeling rater reliability has less room to help, so BBQ’s advantage over simpler models narrows.

The IHQ splits make the rater-quality contribution most visible. BBQ substantially outperforms Crowd-BT and Bayes-BT when lower-quality raters are present, as in IHQ-all and IHQ-unscreened. All methods improve under the screened subset, confirming that rater selection matters. However, Crowd-BT drops on the full dataset even though it models rater quality, likely because crowdsourced raters each provide fewer than 40 comparisons and thus make rater-quality estimation difficult. BBQ maintains strong Top-1 agreement without screening, showing that its Bayesian EM formulation can absorb noisy raters with only a minor accuracy loss.

These results motivate BBQ as a practical evaluation tool rather than only a benchmark improvement. Uniformly high rater quality is difficult to obtain because large-scale crowdsourcing introduces variability, screening is costly, and subjective factors can affect even careful annotators. BBQ provides a principled way to retain partially reliable raters while limiting the damage from noisy contributions.

4.2 SCALING WITH RATERS AND COMPARISONS

The scaling experiment asks which data-collection strategy most supports the BBQ contribution: adding more raters or asking each rater for more comparisons. As suggested by Table 1, datasets

Table 1: Ranking stability of BT-based aggregation methods across datasets. Top rows report Top-1 agreement [%], and bottom rows report Kendall’s Tau. The table highlights the main empirical contribution: BBQ most often identifies the top item, achieves the best overall ranking on more than half of the datasets, and remains second-best on the others.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	<u>33.15</u>
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	<u>98.90</u>	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	0.9180	0.9245	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

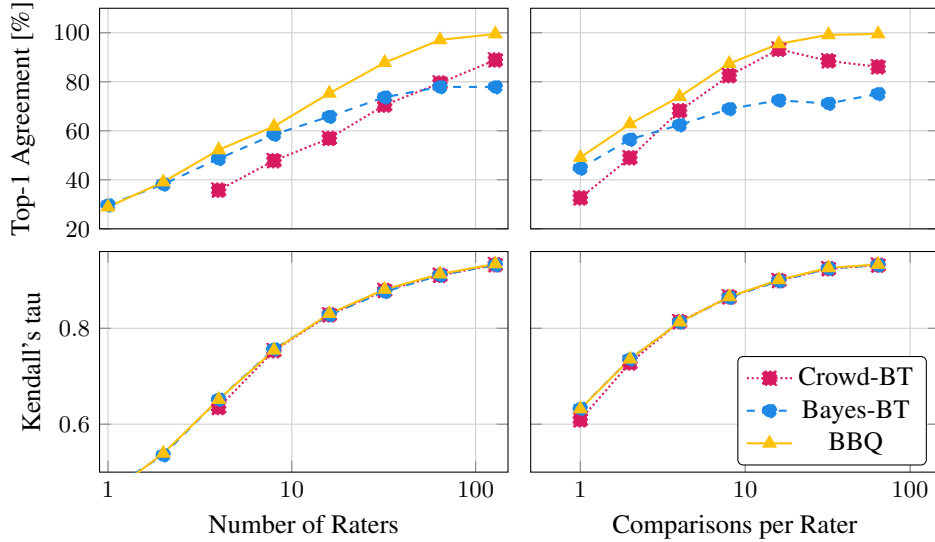


Figure 1: Scaling contribution on IHQ-all for Bradley-Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), BBQ (ours)). **Left:** Performance as the number of raters increases. **Right:** Performance as comparisons per rater increase. More raters or comparisons improve both Top-1 agreement and Kendall’s τ . Top-1 agreement separates the methods most clearly, while Kendall’s τ remains closer across models. Crowd-BT fails to converge with very few raters, making BBQ’s EM-based stability visible. Crowd-BT and BBQ are similar under extreme sparsity, but BBQ surpasses Bayes-BT as either raters or per-rater comparisons increase.

with more comparisons per model generally yield stronger rankings, but those comparisons can be obtained through either axis.

Figure 1 separates these two factors for the three BT variants. The left column varies the number of raters while keeping comparisons per rater at the maximum; the right column varies comparisons per rater while keeping the number of raters at the maximum. Top-1 accuracy reveals the clearest method differences, whereas Kendall’s τ is more similar across models. Crowd-BT fails to converge with only one or two raters, underscoring the practical value of BBQ’s EM updates relative to gradient descent.

Under extreme sparsity, Crowd-BT and BBQ perform similarly because there is too little evidence to distinguish rater qualities. BBQ’s rater-quality contribution becomes more useful once the data include both multiple raters and multiple comparisons per rater. As either dimension grows, BBQ

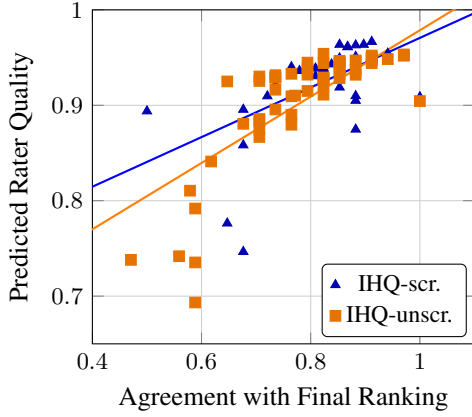


Figure 2: Rater-quality interpretability on IHQ. Each point is one rater, with agreement to the final ranking on the x-axis and BBQ’s predicted rater quality on the y-axis. Triangles denote the screened subset, and squares denote the unscreened subset.

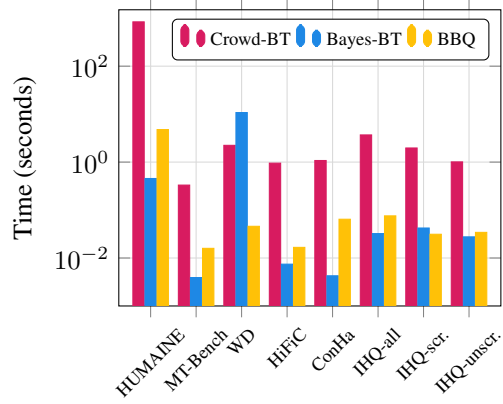


Figure 3: Computational-efficiency contribution measured as average runtime in seconds (log-scale) for one bootstrapped sample across eight datasets. Crowd-BT can require substantial computation time on some datasets, whereas BBQ remains consistently fast.

increasingly outperforms Bayes-BT. Bayes-BT can still surpass Crowd-BT when there are very few raters or little data per rater, because Crowd-BT cannot reliably estimate rater quality in those sparse regimes.

4.3 RATER QUALITY

Figure 2 evaluates whether BBQ’s rater-quality parameters are interpretable rather than merely useful for prediction. Predicted quality increases with agreement to the final ranking, indicating that the model assigns higher influence to raters whose judgments align with the consensus ordering. The Pearson correlation is $r = 0.551$ across 50 raters in the filtered dataset and $r = 0.724$ across 62 raters in the unfiltered dataset.

The unscreened subset contains several raters with both low agreement and low predicted quality. Some IHQ-unscreened raters fall below random-guessing agreement (50%), indicating systematic disagreement with the majority rather than simple noise. This result supports the interpretability claim: BBQ identifies these raters, assigns lower quality scores, and thereby reduces their impact on the final ranking.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 tests the efficiency contribution by reporting the average time needed to process one bootstrapped sample for each dataset. BBQ converges within a few seconds on all datasets, making repeated bootstrap analysis practical. Crowd-BT is slower than BBQ throughout, with especially long runtimes on comparison-heavy datasets such as HUMANE, where convergence takes around 15 minutes. Bayes-BT is slightly faster than BBQ on several datasets but converges slowly on WD. BBQ’s efficiency follows from the closed-form EM updates, which avoid costly gradient tuning while remaining fast on large datasets.

This comparison is conservative for BBQ because Crowd-BT is highly optimized and implemented in C (Kernighan & Ritchie, 1988), whereas BBQ uses plain NumPy without model-specific optimization. A compiled or more heavily vectorized implementation could therefore further reduce BBQ runtime. The observed combination of robustness, stability, and speed supports BBQ as a practical option for large-scale human preference studies and workflows that require repeated bootstrapping.

4.5 CONFIDENCE INTERVALS

The final empirical claim concerns uncertainty: a ranking method should not only order items, but also indicate when apparent differences are statistically meaningful. In a controlled simulation, raters compare two equally skilled items under H_0 , and each model estimates skills and constructs uncertainty intervals as described in Section G. Figure 6 reports the resulting Type I error rates. At the 99% confidence level, the expected error rate is 1%. Crowd-BT and BBQ are close to this target, with slightly lower rates, while Bayes-BT is more conservative at $\sim 0.1\%$. Thus, BBQ preserves well-calibrated uncertainty while adding rater-quality modeling, whereas Bayes-BT underestimates false positives in this experiment.

5 LIMITATIONS

The main limitation follows directly from the contribution: BBQ is most useful when rater quality is heterogeneous. If all raters are uniformly reliable, explicit rater-quality modeling has little additional signal to exploit. Obtaining such homogeneous high-quality rater pools, however, can require substantial cost and effort and may not scale to large studies. BBQ is therefore best viewed as a tool for realistic crowdsourced or partially noisy evaluations, not as a replacement for careful curation when expert raters are available.

The second limitation is input format. BBQ currently handles pairwise comparisons, while many human-evaluation protocols collect ratings, complete rankings, or multi-way choices. Extending the same rater-quality and Bayesian-EM structure to those feedback types would broaden the contribution beyond pairwise studies.

Our empirical validation also has methodological boundaries. Bootstrapping provides a practical measure of ranking stability, but larger-scale studies would further test robustness across evaluation designs. The model assumes comparisons are independent and does not explicitly capture contextual or order effects, both of which can influence human judgments in practice.

Finally, the runtime experiments show favorable scaling, but extremely large item or rater counts could still require optimized or compiled implementations.

6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model that makes rater reliability an explicit part of pairwise-comparison aggregation. The central methodological contribution is a closed-form Expectation-Maximization (EM) algorithm that jointly estimates item skills and individual rater quality, allowing unreliable participants to be downweighted while preserving useful signal from partially reliable raters. This design produces interpretable item and rater parameters, and the EM procedure provides rapid convergence with monotonic likelihood improvement relative to gradient-based alternatives.

The empirical contribution is that BBQ remains stable across diverse language and image preference datasets. It achieves high Top-1 agreement and strong Kendall’s Tau, with the largest gains in noisy or crowdsourced settings where raters vary in attentiveness and diligence. The IHQ experiments show why the rater-quality component matters: BBQ maintains accuracy under unscreened raters, and its predicted rater qualities align with agreement to the final ranking. The Bayesian component further provides uncertainty estimates for item scores, and the simulation study shows that these intervals are well calibrated for assessing statistically significant differences.

Overall, BBQ contributes a practical, interpretable, and generalizable aggregation method for noisy pairwise comparisons. By combining rater-quality modeling, Bayesian uncertainty, and efficient EM inference, it supports more cost-effective and reproducible human studies for evaluating AI-generated content.

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A USAGE OF LLMs

During the preparation of this paper, we used large language models (LLMs) only as auxiliary writing and development tools. ChatGPT, Claude, Gemini, and Grammarly supported spellchecking, phrasing refinement, and text condensation to improve clarity and readability. ChatGPT, Claude, and Cursor also assisted with code analysis, code completion, and visualization generation during the development workflow. All core research ideas, experimental design choices, and interpretations of the results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

This appendix provides the derivation supporting the paper’s algorithmic contribution. We consider R raters comparing K items, where each rater r has quality q_r . With probability q_r , the rater follows the Bradley–Terry model; with probability $1 - q_r$, the rater chooses randomly. The derivation shows how these rater-specific reliabilities can be incorporated into a Bayesian EM (Dempster et al., 1977) framework with closed-form updates.

B.1 MODEL DEFINITION

We use the following notation throughout the derivation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley-Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley-Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley-Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

The Bradley-Terry component assigns probability

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

The rater-quality mixture then gives

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

The latent exponential view used for the EM construction is

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

The complete-data objective decomposes into the following terms:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

The contribution from $P(Z \mid A, \lambda)$ is

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

The contribution from $P(A \mid q)$ is

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

The contribution from the observed comparisons, conditioned on the latent indicators, is

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Combining the three pieces gives the complete-data log-likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

The Bayesian part of BBQ uses conjugate priors: $\lambda_i \sim \Gamma(a, b)$ for each item i and $q_r \sim B(\alpha, \beta)$ for each rater r .

The E-step computes the Q -function, the expectation of the complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

The key responsibility is the posterior probability that an annotation is Bradley-Terry-consistent:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

This responsibility yields the expected sufficient statistics used by the M-step:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Substituting these expectations gives the expected complete-data log-posterior:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

The prior terms contribute

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

The resulting Q -function, which is maximized in the M-step, is

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step completes the algorithmic contribution by maximizing the Q -function with respect to (λ, q) while holding the E-step expectations fixed.

Update for q_r The rater-quality update follows by maximizing Q with respect to q_r , including the Beta prior regularization.

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line is the explicit rater-quality update at iteration t .

Update for λ_i The item-skill update follows by maximizing Q with respect to λ_i , including the Gamma prior regularization.

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)}) \frac{1}{2}}
\end{aligned}$$

where the last line gives the explicit item-skill update at iteration t .

C HYPERPARAMETERS

For Crowd-BT, we use the implementation of Google (2025) with its default hyperparameters, matching the practical baseline used in prior work.

For Bayes-BT and BBQ, the tunable quantities are limited to prior parameters and stopping thresholds. We treat both models as converged when no ELO score changes by more than 1 between two iterations. The ELO score is computed from the skill parameter λ as

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where ELO_SCALE_FACTOR is set to 400.

For both Bayes-BT and BBQ, we use a Gamma prior with shape 5 and rate 0.1 for item skills. For BBQ, the Beta prior on rater quality uses $\alpha = 10$ and $\beta = 2$.

D DATASETS

The dataset selection is designed to test the paper’s contributions across both domain and rater-quality variation. Table 2 summarizes the comparison counts, rater counts, item counts, and per-rater/per-model density.

For natural language, we use HUMAINE (Team, 2025), a large-scale benchmark with over 100k comparisons, and MT-Bench (Zheng et al., 2023), which provides crowd-worker model comparisons on multi-turn dialogues.

For image compression, we include WD (Ballé et al., 2025), a dense expert-labeled dataset with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which focuses on conditional generative models. Together, these datasets vary in scale, number of raters, and rater expertise, giving a broad testbed for ranking stability.

Finally, the IHQ dataset is introduced specifically to expose the rater-quality dimension that BBQ models. It is collected from a user study on CLIC2024 (CLIC, 2025) data through the *Mabyduck* platform and contains two-alternative-forced-choice (2AFC) judgments across 28 generative image compression models. To isolate the effect of rater quality, we provide two subsets: (i) **screened**, containing raters who passed attention and display-quality pre-screening checks, and (ii) **unscreened**, containing all raters. The screened subset approximates a higher-quality rater pool, while the unscreened subset captures the noisier conditions typical of large-scale human evaluation.

Table 2: Dataset summary for the experimental evaluation. The table highlights the variation needed to test BBQ’s contribution: domains differ, comparison density differs, and the IHQ dataset collected on Mabyduck provides screened and unscreened rater pools. Screened subsets include only raters who passed attention and display-quality pre-screening tests, while unscreened subsets include all raters.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

This collection allows us to test both large, relatively clean benchmarks and smaller, noisier settings where the rater-reliability contribution is expected to matter most.

E USER STUDY PLATFORM

All IHQ pairwise comparisons on the CLIC2024 (CLIC, 2025) dataset were collected with the Mabyduck platform (Ltd., 2025). Figure 4 shows the interface used to gather the data. Raters answered the question, “Which image looks more similar to the reference image?”; ties were not allowed, and pairs were selected uniformly at random.

To make rater quality observable in the evaluation, we split IHQ into screened and unscreened subsets. Screened raters passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and they completed training comparisons that reinforced the criterion of selecting the higher-quality image.

Figure 5 shows four examples from the pre-screening process. The first two are standard color-blindness tests in which raters identify the number in each pattern. The final two are shape-detection tests for low-contrast perception: one uses a light gray shape on a white background, and the other uses a dark gray shape on a black background. These checks support the experimental contribution by creating a screened subset whose raters have the visual capabilities needed for reliable image-compression comparisons.

F BOOTSTRAPPING DETAILS

Because BBQ and Crowd-BT explicitly model rater quality, we bootstrap over raters rather than individual comparisons. This preserves each rater’s comparison distribution and therefore evaluates ranking stability under realistic resampling of annotator pools. For all datasets, we use 10,000 bootstrap resamples of raters, except on HUMAINE, where one Crowd-BT bootstrap iteration takes approximately 15 minutes as discussed in Section 4.4. For HUMAINE, we use 1,000 resamples to keep the experiment computationally feasible.

G UNCERTAINTY ESTIMATION DETAILS

Uncertainty estimation is part of BBQ’s contribution because rankings are most useful when evaluators can tell whether observed item differences are statistically meaningful.

Bayes-BT and BBQ quantify uncertainty through the posterior distribution over item skills. Each skill has a Gamma prior updated by the observed pairwise comparisons. We convert posterior credible intervals to the Elo scale so that uncertainty can be interpreted alongside the reported item rankings.

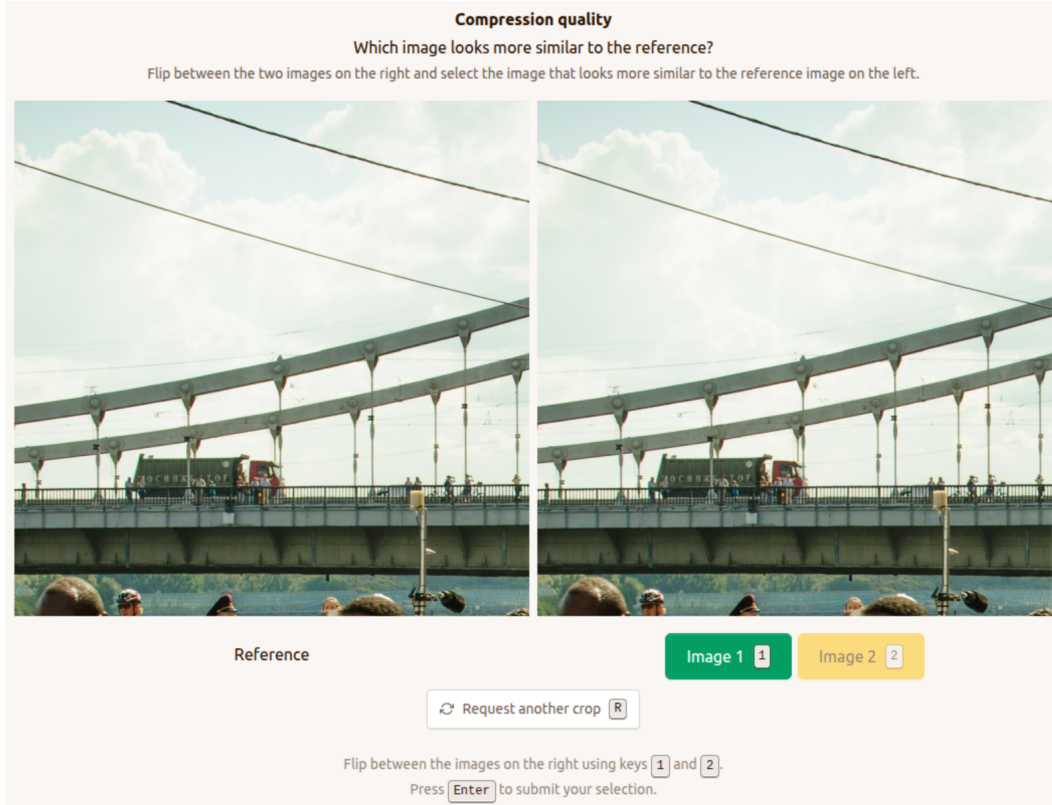


Figure 4: Mabyduck interface used to collect the IHQ pairwise comparisons. The reference image is shown on the left, and the rater selects which of two compressed images on the right is more similar to it.

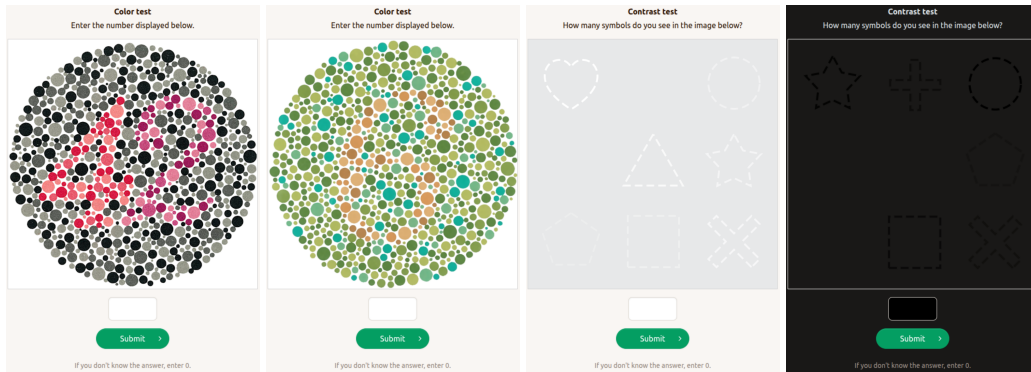


Figure 5: Example rater pre-screening images used to construct the screened IHQ subset. The first two are color-blindness tests requiring number identification, and the last two test sensitivity to low-contrast shapes on light and dark backgrounds.

Crowd-BT is non-Bayesian, so we estimate its uncertainty with a frequentist approximation. We use a second-order Taylor expansion around the maximum likelihood estimate. Inverting the Hessian of the log-likelihood gives a covariance matrix whose diagonal entries are item-level variances. These variances are converted to 99% confidence intervals as

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

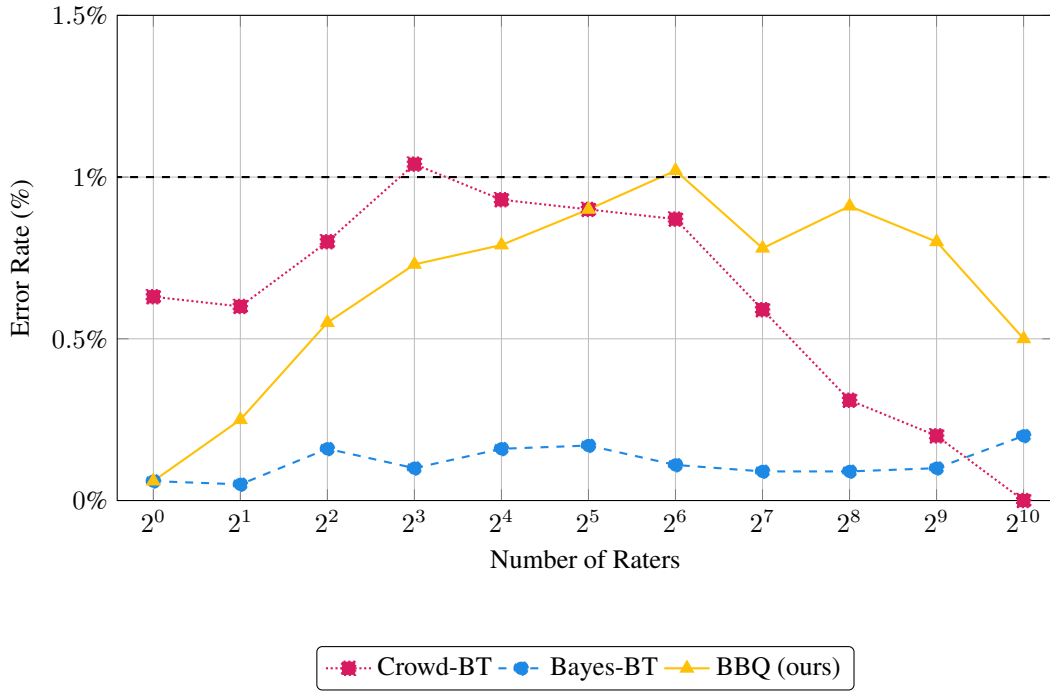


Figure 6: Uncertainty-calibration experiment: Type I error rate as a function of the number of raters. The dashed line marks the expected 1% error rate at the 99% confidence level.

where the constants scale the standard deviation to the 99% confidence level. Narrower likelihood peaks produce smaller intervals and therefore higher certainty, while flatter peaks produce larger intervals and lower certainty.

To test calibration, we run a controlled simulation in which two items have equal skill and therefore a 50–50 win probability. Pairwise outcomes are generated by coin flips, multiple raters are simulated for each trial, and each model estimates Elo scores together with uncertainty intervals.

Formally, the null hypothesis H_0 states that the two items are equally strong. For each trial, we compute the 99% confidence or credible interval for each item skill. A model incorrectly rejects H_0 when the intervals do not overlap, implying a significant item difference even though no true difference exists. The primary metric is the frequency of these incorrect rejections, i.e., the observed Type I error rate.

We perform 10,000 trials, using 50 comparisons per rater, while varying the number of raters to assess how uncertainty estimates scale with the amount of data. This setup allows us to estimate the empirical Type I error rate for each model and compare it against the theoretical expectation of 1% at the 99% confidence level. As shown in Figure 6, Crowd-BT and BBQ exhibit Type I error rates close to the expected 1%, indicating that their uncertainty estimates are well-calibrated. In contrast, Bayes-BT is overly conservative, consistently producing lower error rates than expected, which suggests its credible intervals are not as well-calibrated for significance testing.